Akamill Test plan 2026 - LLDC, platform service and data flows

- Current Test cases on JIRA
- Early discussions
  - LLDC or DDC-Role test coverage
  - Akamill as a service - functional tests
  - Akamill - e2e data flows (xforms, graphs, lookups)
- Test cases - Reviewed must be added to Jira TC
  - Existing Tables: Review with Vadim
  - LLDC Leader
  - Testing e2e data flows
    - BTS - passthru - flow
    - Beacon - ETP AnswerX - flow
    - Ghost rolls - pass through - flow
    - Beacon - SIEM - flow
    - akamillbridge - cedexis DLR passthrough - flow

Akamill User Guide

Current Test cases on JIRA

Key	Summary	T	Created	Status	Resolution	Labels
DDCSQA-6348	[GHost][APISEC] To Validate Akamill Processing For APISEC Via Query2		Apr 30, 2026 14:57	OPEN	Unresolved	Akamill DDC GovDDC Release TO_BE_CLEANED
DDCSQA-6347	[GHost][test-sink] To Validate Akamill Processing For Ghostrolls Via Query2		Apr 30, 2026 14:42	IN PROGRESS	Unresolved	Akamill DDC GovDDC Release TO_BE_CLEANED
DDCSQA-6345	[Smoosh/ComputeL2][ddc-bz-cmpl-in] To Validate Akamill Processing For Completeness Tracking Via Query2 - HTS		Apr 30, 2026 11:41	IN PROGRESS	Unresolved	Akamill DDC GovDDC Release TO_BE_CLEANED
DDCSQA-6344	[Q/PSplit/ComputeL1][ddc-bz-cmpl-in] To Validate Akamill Processing For Completeness Tracking Via Query2 - BZS		Apr 30, 2026 11:39	IN PROGRESS	Unresolved	Akamill DDC GovDDC Release TO_BE_CLEANED
DDCSQA-6212	[ESSL-APISEC][APISEC-linode] To Validate APISEC Akamill E2E leucent flow		Apr 02, 2026 15:05	REVIEW	Unresolved	Akamill DDC Release TO_BE_CLEANED
DDCSQA-6211	[ESSL-APISEC][APISEC-non-linode] To Validate APISEC Akamill E2E eucentral1 flow		Apr 02, 2026 15:05	REVIEW	Unresolved	Akamill DDC Release TO_BE_CLEANED
DDCSQA-6210	[ESSL-APISEC][APISEC] To Validate APISEC Akamill E2E mecentral1 flow		Apr 02, 2026 14:59	REVIEW	Unresolved	Akamill DDC Release TO_BE_CLEANED
DDCSQA-6209	[FF-APISEC][APISEC-linode] To Validate APISEC Akamill E2E leucent flow		Apr 02, 2026 14:58	REVIEW	Unresolved	Akamill DDC Release TO_BE_CLEANED

DDCS QA-6147	[FF-APISEC][APISEC-linode] To Validate APISEC Akamill E2E eucentral1 flow		Mar 27, 2026 05:57	REVIEW	Unresolved	Akamill DDC Release TO_BE_CLEANED
DDCS QA-6127	To Verify Failover scenario		Mar 24, 2026 10:46	OPEN	Unresolved	Akamill DDC Release TO_BE_CLEANED
DDCS QA-6102	[GHost][ddc-bz-cmpl-in] To Validate Akamill Processing For Completeness Tracking Via Query2 - BTS		Mar 19, 2026 08:02	IN PROGRESS	Unresolved	Akamill DDC GovDDC Release TO_BE_CLEANED
DDCS QA-6084	To Verify Retry mechanism and persistent storage		Mar 18, 2026 06:44	OPEN	Unresolved	Akamill DDC Release TO_BE_CLEANED
DDCS QA-5978	Latency/Performance Test for Akamill		Mar 14, 2026 07:59	OPEN	Unresolved	Akamill
DDCS QA-5964	To verify service manager actions for akamillp service		Mar 13, 2026 12:44	QA ACCEPTED AUTOMATION	Unresolved	Akamill DDC Release TO_BE_CLEANED
DDCS QA-5962	To verify service manager actions for akamilld service		Mar 13, 2026 12:40	QA ACCEPTED AUTOMATION	Unresolved	Akamill DDC GovDDC Release TO_BE_CLEANED
DDCS QA-5939	[DataFlow] Akamill - Ghost rolls Passthrough Flow Validation		Mar 11, 2026 09:54	IN PROGRESS	Unresolved	Akamill DDC Release TO_BE_CLEANED
DDCS QA-5826	[Ghost][BOT] To Verify Akamill E2E Flow - Legacy (DLR endpoint)		Feb 23, 2026 13:39	OPEN	Unresolved	Akamill DDC Release TO_BE_CLEANED
DDCS QA-5825	[GHost/QPSplit/Smoosh][ddc-bz-cmpl-in] To Verify Akamill Flow - Completeness Tracking		Feb 23, 2026 13:34	QA ACCEPTED AUTOMATION	Unresolved	Akamill DDC GovDDC Release TO_BE_CLEANED
DDCS QA-5824	[GHost][CSI] To Verify CSI_AZ_WAF_TRIG E2E Flow		Feb 23, 2026 13:29	QA ACCEPTED AUTOMATION	Unresolved	Akamill DDC Release TO_BE_CLEANED
DDCS QA-5823	To verify service manager actions for lldc_le service		Feb 23, 2026 13:27	QA ACCEPTED AUTOMATION	Unresolved	Akamill DDC GovDDC Release TO_BE_CLEANED

Showing 20 out of 52 issues

Early discussions

LLDC or DDC-Role test coverage

LLDC - Low Latency Data Collection

- [x] Install tests
- [x] FE, BE configurations
- [x] Servicemgr tests for akamillp, akamilld, ddc_nginx, ddc_dlr_xfer
- [ ]

similar to prod checklist DDCPRODCTS-2170 - Getting issue details... STATUS

- SOURCES / checklisting-scripts / **pychecklist_scripts / ddc_akamill**
  master
  ```
  100644 blob 73efbd05205d08ac10699d31f05866f50cd9248f        .gitignore
  100644 blob a9b11c2c0180a39a96db4b1f330b8678f03fe77f        README.md
  100644 blob 38329bafb3f90b52dd9b9df2465420f11fff0aca        config.yaml
  100644 blob 2c1f17b5ced228e1501850dc0e30d86901c0b268        ddc_akamill.py
  100644 blob 08e0878265c9d284487b1c019fb5ca83072fe3a6        group_checks.py
  100644 blob 707691da39525501667cd1c5861e89a57f6965c8        machine_checks.py
  100644 blob 99f1946d56d48eacd7858471de9ef30f23106375        utils.py
  ```
  - rolls, coredumps, service status, certs

- [ ] Monitoring
  - Process rolls
  - Queue process rolls
  - AMS alerts
  - dmesg
  - grafana dashboards
  - performance tests
  - latency tests

90% of LLDC servers are (manually set configs) configured to be FEs, and rest are BEs. They can switch roles depending on the load.

Akamill as a service - functional tests

- [x] AuthN (authentication http/https/mtls)
  - DDCSQA-5574 - Getting issue details... STATUS
- [ ] AuthZ (authorization)
  - Data perms
- [x] Persistent Retry
- [ ] LLDC sender
- [x] lldc tables
- [x] lldc queues
- [ ] handlers

Akamill - e2e data flows (xforms, graphs, lookups)

- Ingest - Types of flows
  - [x] Non-native flows – https post (no akamill client)
  - [x] Native – by akamill client on networks
- Data Source
  - [x] Only one of data providers need be tested (say FF or ESSL, or cover across various tests)
  - [x] edge-akamilld HTTPS POST to LLDC
    - FF to LLDC covered by e2e test
  - [x] direct HTTP(S) POST to LLDC
    - ESSL to LLDC covered by e2e test
  - [x]

beacons (legacy)
    - DDCSQA-1724 - HTTP / HTTPS / HTTPS+mTLS
    - WAF DLR for Gov
- Data xform (processing)
    - [ ] (non native ingest flow coverage) CSI + WAF DLR
    - [ ] ~~BOSS beacons (DNA) - this may migrate soon per Max~~
        - ~~no media analytics~~
    - [ ] etp, answerx
    - [ ] MSL4
    - [x] MSFT-NRT
    - [ ] ~~Edge workers (will be shutdown soon)~~
    - [ ] with message receivers on DAS
- Data - Destination flow
    - FE to sink
    - FE to BE to sinks
    - FE to LLDC sender to sink
- Communication with Hilbert and other systems for completeness tracking

Akamill service tests - nginx accepts data via HTTP, HTTPS and mTLS POST

- ESSL sends HTTPS-POST with data to LLDC/Akamill
    - script by Phil executes a curl
    - response - status 200OK check on sender side
    - (will be covered again in e2e tests as well, but covering separately as a sanity test)
- Akamill/LLDC receives data (where destn is /beacons/csi ,by nginx, on LLDC)
    - will be covered in e2e flow tests
- optional LLDC FE sends data HTTPS-POST to LLDC-BE
    - if FE works, BE also should work. No need for separate test
    - anyways this flow will be covered in e2e test
- LLDC-BE (or FE) , after processing, does HTTPS-POST towards destination (CSI sink)
    - uploader test - will be covered as part of e2e tests

Test cases - Reviewed must be added to Jira TC

	SME	Scenario	Description	Detailed Steps	Comments	TC on Jira (new/update)
1	Yuvarani	Network config - Installing Akamill /LLDC servers	Covered as part of install tests in MTP		[x] Reviewed - Vadim	[x] Done

2	Miloslaw	akamill FE and BE configurations	Check (2 only akamill regions)
			
			• at the least 1 server is FE, and at the least 1 is BE (or as per configurations)
			  ○ check nginx log to confirm?
			  ○ any query table? ddc_beacon_receiver_beacon_stats?
			
			"select * from ddc_beacon_receiver_beacon_stats where feed = 'msft_nrt_asgard'"
			
			machineip start_time end_time feed xferred failed dropped unparsed_kept unparsed_dropped_routing unparsed_dropped_xform pending incoming
				check table lldc_topology - backends will have rows	☑ Reviewed - Vadim	☑ Done
						
						DDCSQA-5817 - Getting issue details... STATUS
						
						DDCSQA-5818 - Getting issue details... STATUS

3	Miloslaw	lldc_le service	leader election in a region - Leader publishes query2 info for region. (election is by a mapper function)	table - lldc_region_leader - verify 1 leader exception - rarely more than 1 leader could be elected debug , check with dev-sme if consistent. Leader publishes query2 stats. any other functions for a leader? No.	☑ Reviewed - Vadim	☑ Done **DDCSQA-5819** - Getting issue details... STATUS

4	Miloslaw	akamilld service (akamillp is obsolete - no testing needed) akamill service group	ServiceMgr tests in MTP - stickystop/stickysuspend /restart - check for rolls (stop/start/suspend) service group level tests needed. (akamillp - how is it used in prod - low priority)	Raghu - At the service level group also we should perform test /a/sbin/akamai_run check lldc lldc (group) : Running OK lldc_le : Running OK ddc_nginx : Running OK ddc_dlr_xfer : Running OK /a/sbin/akamai_run check akamill akamill (group) : Running OK akamilld : Running OK akamillp : Running OK Vadim - akamillp is not directly used as a service; however, it is referenced by some Query2 views (like ddcRoles), so it's "used" in that sense.	☑ Reviewed - Vadim	☑ Done DDCSQA-5820 - Getting issue details... STATUS
5	Miloslaw	ddc_nginx service lldc service group	ServiceMgr tests in MTP - stickystop/stickysuspend /restart - check for rolls suspend at lldc only nginx suspends which will stop receiving data.		☑ Reviewed - Vadim	☑ Done DDCSQA-5821 - Getting issue details... STATUS
6	Miloslaw	ddc_dlr_xfer service lldc service group	ServiceMgr tests in MTP - stickystop/stickysuspend /restart - check for rolls		☑ Reviewed - Vadim	☑ Done DDCSQA-5822 - Getting issue details... STATUS
7	Miloslaw	lldc_le service lldc service group	ServiceMgr tests in MTP - stickystop/stickysuspend /restart - check for rolls		☑ Reviewed - Vadim	☑ Done DDCSQA-5823 - Getting issue details... STATUS

8	-	sub/ services will not be covered in SQA	• ddc_dlr_xfer - unit tests devQA - review again		☑ Reviewed - Vadim	n/a
9	Aditya /Yuvarani	cover prod checklist	get details from Vadim. is that script executable in testnet. ddc_nginx, akamill, beaconreceiver	Vadim - All checklist scripts are in https://git.source.akamai.com/projects/SOURCES/repos/checklisting-scripts/browse/pychecklist_scripts : • https://git.source.akamai.com/projects/SOURCES/repos/checklisting-scripts/browse/pychecklist_scripts/ddc_nginx • https://git.source.akamai.com/projects/SOURCES/repos/checklisting-scripts/browse/pychecklist_scripts/ddc_beacon_receiver • https://git.source.akamai.com/projects/SOURCES/repos/checklisting-scripts/browse/pychecklist_scripts/ddc_akamill • technically, there's also https://git.source.akamai.com/projects/SOURCES/repos/checklisting-scripts/browse/pychecklist_scripts/akamill_lua_xform, but it is just a wrapper around the ddc_akamill one.	☑ Reviewed - Vadim	☐ Done

10	Yuvarani	(process) Rolls	roll monitor		☑ Reviewed - Vadim	TC exists for general rollmon
11	Yuvarani	LLDC Queue rolls	(roll monitor for queues) Queue Rolls are monitored with help of table - ddc_dlr_process_rolls Existing test case - **DDCSQA-3587** - Getting issue details... STATUS		☑ Reviewed - Vadim	☑ Done
12	Yuvarani	AMS alerts	alerts monitored for a release		☐ Reviewed - Vadim	TC exists for general alert monitoring
13	Miloslaw	dmesg	dmesg are monitored for duration of a release		☑ Reviewed - Vadim	TC exists for general dmesg checks

14	Raghu	Latency measurements of Akamill	Benchmarking latency - (with synthetic data, locally ingested, destn: localhost/dnsresolution) • latency for passthru - (FE only in path) - ○ 3 ingest types • latency for a fixed graph (FE and BE in path) - Check for degradation in every release with Akamill changes. (Consider other influencers. Network latency may influence if BE is in path?) (prod issue) Burst of traffic - akamill queue should be to handle burst, if within queue buffer limit. alert 31834 - to detect akamill data loss due to resource exhaustion		☑ Reviewed - Vadim ☐ Review again w/Vadim when test has detailed instructions, especially monitoring graphs.	☑ Testcase created - DD CSQA-5978 ☐ Details - TBD

15	Raghu	AuthN	• data by (HTTP) , HTTPS, mTLS
			• internal hops - FE to BE hop is currently over port 8482, w/o using a client cert
			
			the FE->BE hop done by dataflows that arrive to the DLR endpont (/beacons/...) use the /enqueue endpoint.
			
			The hop that occurs in the akamill graph (as represented by <queue> vertices mentioning ' /akamill/fwdq') use the /akamill endpoint instead.
			
			• (only 8484 for GovDDC)
			• **cert rotations** - decide if this needs to be in regression
			  ○ maybe only when akamill changes, not every release
			  ○ keep the tests ready
			• akamill-ingest - does it authenticate in any way? this is used for local ingestion, by other services.
			• include negative tests - wrong cert presented
			• liveness tests 8482, 8484, 8481(important per Xiaoping) - can test on FE or BE. only FE is tested for liveness for now, by gtm. nginx access logs. Periodic - 1 second?
			  ○ negative test for liveness
			  ○ disk full
					☑ Review with Vadim
					☑ Vadim to confirm on FE to BE
					☑ Vadim to share more about ports
						☐ cert rotation
						☐ ports liveness tests
						☐ DataFlow TC - ?? - add a note in those test cases

16	Damian	AuthZ	root/logger/testgroup /akamai (gov has other users - already covered in tests in GovDDC) tests at high level - • a user can read/write files - service logs, temp files • a user can stop/start /restart services • a user can read/write config files, access binaries • tmpfs queues and retry queues will keep backing up if access is wronged. q2 table - Vadim will share	Need details from Vadim	☑ Reviewed by Vadim ☐ Contact Vadim for specifics	☐ Done
17		FLOW2 - Test coverage for flow - completeness - Q/P (service akamilld) Hilbert FLOW3 - ComputeL1 (akamilld) DDC Akamill Appbattery		Flow 2 (Hilbert - bts covers this) and Flow 3 are covered in data flow list below		n/a

18	Raghavendra Thalinki	FLOW8 - Test LLDC **retry** mechanism (transient and persistent) - is LLDC sender covered in this?	**csi_linode_cln_traf_perf_test.feed.yaml**
			
			`routing:`
			`  tags:`
			`  - csi_az_cln_traf`
			`aggregation:`
			`  threshold_seconds: 2`
			`  threshold_bytes: 2m`
			`retry:`
			`  enabled: true`
			`  persistent_storage: true`
				(Vadim) persistent retries (i.e., ensure lldc_rexmit_sender works ok)
				
				-- covered in data flow test list
						covered in data flow test list below

19	Aditya Mahalwar	~~Test coverage for flow - ??~~ ~~sample usecase – Edge Worker – low priority?~~	~~Verify the flow for Edge worker pipelines flowing through akamill.~~ ~~**1. Pipeline:** ew_subrequests~~ ~~Verification of ew_subrequests Pipeline~~ ~~**2. Pipeline:** ew_init~~ ~~Migration of ew_init pipeline~~ ~~**3. Pipeline:** ew_event_handler~~ ~~Migration of ew_event_handler Pipeline~~	~~Flow 1:~~ ~~Ghost Akamill (FEBE) Smoosh (Flare) Azure Asgard Pierun (this is covered in MSFT flow below)~~ ~~Flow 2: multiple destn?~~ ~~Ghost Akamill (FEBE) DAS~~	EW flow will not be supported soon	
20	Aditya Mahalwar	Test coverage for flow - ?? usecase - ?? Messager-receiver - DAS Sink	Flow 6: Ghost Akamill DAS Messager-receiver - DAS Sink Verify the Pipelines which sends data to DAS as sink. root@198.19.9.230:/a/share/logger/mdt/ddcdir.lldc.sinks# grep -l das * edgeworkers_event_handler_flare_dfw.connection.yaml edgeworkers_event_handler_flare_ewr.connection.yaml edgeworkers_ew_init_flare_ewr.connection.yaml edgeworkers_ew_subrequests_flare_dfw.connection.yaml edgeworkers_ew_subrequests_flare_ewr.connection.yaml edgeworkers_ew_subrequests_flare_lax.connection.yaml	Flow 6: Ghost Akamill DAS is this data flow needed? (see if already covered in data flow list below)		

			edgeworkers_ew_subrequests_flare_ord.connection.yaml rapid_api_key_tr_dfw.connection.yaml rapid_api_key_tr_ewr.connection.yaml rapid_api_throttling_v2_dfw.connection.yaml rapid_api_throttling_v2_ewr.connection.yaml rapid_cache_hit_dfw.connection.yaml rapid_cache_hit_ewr.connection.yaml rapid_cache_miss_dfw.connection.yaml rapid_cache_miss_ewr.connection.yaml rapid_endpoint_req_stats_by_account_v2_dfw.connection.yaml rapid_endpoint_req_stats_by_account_v2_ewr.connection.yaml rapid_endpoint_req_stats_by_endpoint_v2_ewr.connection.yaml rapid_fwd_turn_dfw.connection.yaml rapid_fwd_turn_ewr.connection.yaml rapid_jwt_invalid_dfw.connection.yaml rapid_jwt_invalid_ewr.connection.yaml rapid_jwt_valid_dfw.connection.yaml rapid_jwt_valid_ewr.connection.yaml				

			rapid_oauth_invalid_dfw.connection.yaml rapid_oauth_invalid_ewr.connection.yaml rapid_oauth_valid_dfw.connection.yaml rapid_oauth_valid_ewr.connection.yaml rapid_orig_fail_dfw.connection.yaml rapid_orig_fail_ewr.connection.yaml rapid_orig_fail_v2_dfw.connection.yaml rapid_orig_fail_v2_ewr.connection.yaml rapid_quota_deny_dfw.connection.yaml rapid_quota_stat_dfw.connection.yaml rapid_quota_stat_ewr.connection.yaml rapid_quota_stat_v2_dfw.connection.yaml rapid_quota_stat_v2_ewr.connection.yaml rapid_req_deny_dfw.connection.yaml rapid_req_deny_ewr.connection.yaml rapid_req_err_dfw.connection.yaml rapid_req_err_ewr.connection.yaml rapid_req_succ_dfw.connection.yaml rapid_req_succ_ewr.connection.yaml rapid_rsp_turn_dfw.connection.yaml				

			rapid_rsp_turn_ewr. connection.yaml rapid_tfc_rsp_dfw. connection.yaml rapid_tfc_rsp_ewr. connection.yaml waf_exceptions. connection.yaml waf_exceptions_chicago. connection.yaml			

21	Yuvarani Jagadesan	LLDC Table Verification	Based on DDC-akamill checklist in DDC doc, below tables are planned for DDC-Release verification. 1. akamill_basic_stats - Checks the incoming streams, xforms received, processed and dropped are verified. 2. lldc_frontend_stats - Checks the FE machines response(200), hits and bytes are populated 3. lldc_outgoing_byte_stats - Checks the BE/Sink status failed or dropped shouldn't be seen 4. lldc_outgoing_retry_byte_stats - Checks the retry for outgoing bytes status. Empty rows should be seen. (In SQA currently we have rows with sink ip as 0.0.0.0 for few sinks so there are few rows seen) ddc_dlr_process_rolls - Being tested in current DDC Release as part of Akamill Queue Roll Monitor. Discuss existing tables listed below in this doc with Vadim Yuvarani Jagadesan	https://track.akamai.com/jira/browse/DDCSQA-5650		☐ Done - list of tables reviewed by Vadim ☐ Done - TC has steps for all tables suggested by Vadim

22	Yuvarani Jagadesan	LLDC DDCwatchdog Handlers Check	Grafana Dashboard in Section : "By release: ddcwatchdog handler failures" https://grafana.akam.ai/org293/dashboard/db/lldc-process-rolls-shared?refresh=15m
			
			Query Used
			
			`SELECT h.machineip machineip, swlabel, handler, sum(ntried), sum(nfailed), sum (nsignaled), sum(ncored) FROM ddc_ddcwatchdog_svc_status_hour h,`
			`  installinfo ii`
			`WHERE h.machineip IN (select machineip from whoruns where lldc=1)`
			`  AND h.machineip NOT IN (`
			`  SELECT s.machineip`
			`  FROM stickyreasons s, installinfo ii`
			`  WHERE service="lldc"`
			`  AND s.machineip = ii.machineip`
			`  AND timeoccured > installdate`
			`  )`
			`  AND handler LIKE "lldc%" and nfailed+nsignaled+ncored > 0`
			`  AND handler NOT IN ("lldc_validate_region_load_data", "lldc_validate_region_load_data")`
			`  AND h.machineip = ii.machineip`
			`  AND timestamp > installdate`
			`GROUP by 1,2,3 order by handler, swlabel, machineip`
				https://track.akamai.com/jira/browse/DDCSQA-5595		☑ Done

23	Yuvarani	Dashboards for monitoring	DDCSQA-5501 - Dashboards for monitoring
			
			Review Done
			
			• LLDC Process Resource Usage Grafana Dashboard - Checking **CPU, RSS and vsize of Frontend and Backend : By Release** only for the below commands:
			  ○ akamilld
			  ○ ddc_beacon_receiver
			  ○ ddc_nginx
			  ○ ddc_dlr_forwarder
			  ○ lldc_le
			  ○ lldc_log_scanner
			• LLDC System Resource Usage Grafana Dashboard
			  ○ Checking the incoming bytes to akamill and no drops in the bytes (**Notes Section : Akamill SV bytes /s**)
			  ○ Checking the incoming bytes to akamill and no Drops in the message (**Notes Section : Akamill SV msg/s**)
			  ○ Checking the graphs for no spikes or drops in section "**System Resource Usage: by Release"**
			  ○ Checking the tmfs folders are not dropped or spiked in section **"tmpfs usage by Release"**
			
			Test Case - DDCSQA-5501
						☑ Done

24	-	-	Understanding of DevQA coverage - expected to have full coverage of component level functionality. Unit /functional tests.	get info from Vadim /Jane		
25			Jeff - 1) Could we add testing for 12 and sixteen threads (as soon as possible) on ECP16 & X8 machines please see https://track.akamai.com/jira/browse/DXRE-1379 & https://track.akamai.com/jira/browse/DDCINFRA-4341	Review w/Vadim		
26			Jeff - #2) Can we attempt to reproduce the 'qmgr.cc: 865: writebuffer: Qmanager is not active 0x20' failure seen here https://track.akamai.com/jira/browse/DXSTRMESC-463	Review w/Vadim		
27			Jeff - #3) Can we add a test which increase the retry queue period with a dead sink and see the roll behavior when we exhaust memory.	Review w/Vadim		
28			Jeff - #4) Can we adjust the nginx listen queues down (to 10 or so) to cause listen queue overflow and see the steady state behavior	Review w/Vadim		
29			Jeff - #5) Can we restrict the size of the /tmpfs partition to see the behavior when it is exhausted.	Review w/Vadim		
30	Raghu					☐ Done

Existing Tables: Review with Vadim

* ddc_dlr_queues
* akamill_basic_stats
akamill_basic_stats_kafka_sender
akamill_bridge_dlr_drops
akamill_bridge_feeds
* akamill_detailed_stats - to be covered in e2e stats, grafana
akamill_detailed_stats_kafka_sender
akamill_ingestor_stats
akamill_inotify_stats
akamill_kafka_sender_stats
akamill_proxy_stats
* akamill_sender_stats
* akamill_stream_vertex_stats
* akamill_xform_vertex_stats

* lldc_bad_sink_conf - for negtaive tests
lldc_bridge_sinks
lldc_frontend_batch_stats
lldc_frontend_incoming_error_stats
lldc_frontend_incoming_stats
lldc_frontend_stats *
lldc_frontend_url_stats *
lldc_gtm_region_info
lldc_instances *
lldc_outgoing_beacon_stats
* lldc_outgoing_byte_stats
lldc_outgoing_latency_stats * beacon receiver only
lldc_outgoing_request_stats
lldc_outgoing_retry_beacon_stats
lldc_outgoing_retry_byte_stats *
lldc_outgoing_retry_request_stats *
lldc_owner_map
lldc_owners
lldc_projection_map
lldc_projections
lldc_region_leader * - covered in DDCSQA-5819 - Getting issue details... STATUS
lldc_region_load.
lldc_sink_info
lldc_topology

lldc_uploader_stats

ddc_beacon_receiver_beacon_stats
ddc_beacon_receiver_beacon_track_stats
ddc_beacon_receiver_byte_stats
ddc_beacon_receiver_byte_track_stats
ddc_beacon_receiver_latency_stats
ddc_beacon_receiver_request_stats

LLDC Leader

1. region leader info.
The region leader info is published to the lldc_region_leader table (link shows the table doc).
On each region leader, a file exists that indicates leadership:
- /a/var/run/ddc_dlr_xfer/lldc_le.region_leader
On every LLDC machine, the IP of the region leader can be found in
- /a/var/run/ddc_dlr_xfer/lldc_le.region_leader_ip

As I mentioned, it is primarily used for for query publishing
(each region publishes some information from the leader, including the following tables:
lldc_topology
lldc_gtm_region_info
akamill_bridge_feeds
lldc_bridge_sinks
lldc_sink_info
lldc_owners, lldc_owner_map
lldc_projections, lldc_projection_map
I don't believe it's currently used for anything else.

Testing e2e data flows

	Data flows **(business case)**	**Data provider**	**Ingest Type** **(native /non-native)**	**Akamill** **Processing** **(passthru /..)**	**Sink for Akamill** **(not all sinks need to be covered)**	**Destination** **(for e2e flow) (dest coverage is not reqd)**	**e2e Data path** **(including akamill FE, BE)**	**Test steps**
1	☑ (STG) CSI WAF DLR	ESSL	non-native - http /https? post to DDC (Role LLDC, akamilld) akamill bridge endpoint	passthru			directly goes to 2 sinks (uncompressed). to send compressed data to other 2 sinks data is also copied and send to akamilld beacon-receiver, and then processed? Flow : DLR /beacon : Ghost Akamill FE sink	

2	☑ (STG) CSI WAF DLR	Gov ESSL	non-native - https post to GovDDC (Role Akamill, akamilld)	passthru			FLOW5 - DLR /beacon : Ghost Akamill FE Akamill BE sink	
3	☑ Ghost rolls	FF, ESSL, ..	akamill native endpoint	FE-only, no BE. Passthrough			(Vadim) simple feeds arriving via akamill native endpoint (/akamill): - FE-only, no BE: ddc:: akamillp::ghost:: ghostroll::0 (ddcdir.akamill. graph.ddc /ddc_roll.xml) check ddcdir. akamill.graph. freeflow (FLOW 1)	

4	☑ Hilbert - bts/hts /bzs		akamill native endpoint	FE+BE			(Vadim) - simple feeds arriving via akamill native endpoint (/akamill): - FE+BE: ddc:: akamillp::ghost:: bts::0 (ddcdir.akamill. graph.ddc/bts.xml) How does FE send to BE, which endpoint? Vadim to confirm. For akamill endpoint, identifier is sent along with data (its not in url, as in dlr case) (FLOW2 - Test coverage for flow - completeness - Q/P (service akamilld) Hilbert) (FLOW4 - SMOOSH Akamill (for Hilbert))	

5	☑ GovDDC to DDC for Hilbert /completeness						FLOW3 - ComputeL1 (akamilld) DDC Akamill Appbattery	

6	-		~~akamill-bridge-endpoint~~	DO NOT TEST - this one is not receiving data in prod, but config still exists. ~~This is suggested as its simplest case for akamill-bridge-endpoint. This is covered in waf-dlr.~~			~~(Vadim)~~ ~~feed arriving via akamill-bridge-endpoint~~ ~~- /akamillbridge/cedexis~~ ~~- (ddcdir.akamill.graph.ddc/cedexis.xml + ddcdir.ddc_nginx.conf/akamill_bridge.conf)~~ ~~—~~	
7	☑ legacy (DLR) endpoint		legacy (DLR) endpoint.				(Vadim) feed arriving via legacy (DLR) endpoint: ~~- /beacons/etp_ax_packet~~ ~~- (ddcdir.lldc.sinks/etp_ax_packet_linode_prod_dc_1_us_*.*.yaml)~~ doesnot go to akamilld. gets directly sent. Used bot related flow instead of answerx suggested above	

8	☑ SIEM flow - from akamilld to ddc_beacon_receiver to akamillbridge		ddc_beacon_receiver				(Vadim) feed bridged from akamilld to ddc_beacon_receiver (the LLDC sender) (and also is internally routed to akamillbridge) /beacon/siem_v1 (this one is a bit more involved; ddcdir.lldc.sinks/siemv1_*.yaml and ddcdir.akamill.graph.ddc/siem.xml) akamilld sends data to ddc-beacon-receiver.	
9	☑ MSFT NRT	FF	beacons (legacy) native - FF (akamilld?) to DDC (Role LLDC, akamilld)	FE to BE			FLOW7 Ghost AKAMILL SMOOSH AZURE Verify Akamill can receive, process, and send beacons to DDC (SmooshClassic Uploaders) for upload to Azure	Verify that **Akamill as a platform** is able to:

a.

b.

c.

d.

10	☐ Failover						(Vadim) FE/BE hop (normal + failover to FE) (i.e., ensure ddc_dlr_forwarder works ok) cover these - • legacy (DLR) endpoint (see test above) can be used. shut BE down, and send data to FE. • akamill endpoint - if BE is down, it should failover to FE	

11	☐ Retry		Retries				check with Vadim for doc on yaml/configs.
							
							csi_linode_cln_traf_perf_test.feed.yaml
							
							`routing:`
							`  tags:`
							`  - csi_az_cln_traf`
							`aggregation:`
							
							`threshold_seconds: 2`
							
							`threshold_bytes: 2m`
							`retry:`
							`  enabled: true`
							
							`persistent_storage: true`
								• retris by bea on rece ver. pers ster stor ge is not.
								• retris by bea on rece ver. pers ster stor ge is enaed

								• retris by akai illd - tran ent (try limit d num of retri s, and ther drop . us hos ame poir s to 2 IPs, and mak one dow . • max ge - expi atio from pers ster que e. (Vadim) persiste t retries (i.e., ensure lldc_rex mit_ser er work ok)

								FLOW8 - Test LLDC **ry** mechar sm (transie and persiste t) - is LLDC sender covered in this?
12	☐ API sec						Check with Xiaoping for details APISEC akamill path EPIC - **DDCINFRA-4870** - Getting issue details... STATUS DDCIRC-1373/DDCIRC-1374/DDCIRC-1375/DDCIRC-1376 should be pushed as a group (order doesn't matter) to enable the apisec new data path. FLOW5 - akamill end point : Ghost Akamill FE Akamill BE sink. To be confirmed if this is the flow. Need graph diagram from Vadim.	
13	☐ ETP-AnswerX							Verify the E2E flow for ETP pipeline

							ETP - is this needed? Is ETP-answerX pipelines are same as bounce ETP ? check if already covered in previous rows Aditya Mahalwar Yeah, I check this is already covered under TC : DDCSQA-5826	Pipeline found for ETP blr-mp7er:ddcdir.lldc.sinks amahalwa$ pwd /Users/amahawa/ddc_pc_amahawa_blr-mp7er/metada/ump/ddcdir.lldc.sink blr-mp7er:ddcdir.lldc.sinks amahalwa$ cat sink_index.yaml	grep etp - etp_a_packet linode_ od_dc_ _us_iad - etp_a_packet linode_ od_dc_ _us_lax - etp_a fh_linod _us_iad - etp_a fh_linod _us_lax

								- etp_e aa_dp_ node_u _iad - etp_e aa_dp_ node_u _lax - etp_r evada_ node_u _iad - etp_r evada_ node_u _lax - etp_s nkhole_ node_u _iad - etp_s nkhole_ node_u _lax blr- mp7er: ddcdir. lldc. sinks amahal a$
14	☐ DAS flow is this needed Raghavendra Thalinki ?							Messag r- receive - DAS Sink Flow 6: Ghost Akamill DAS Messag r- receive - DAS Sink

								Verify the Pipeline which sends data to DAS as sink. root@18.19.9.230:/a/share/logger/mdt/ddcdir.lldc.sinks# grep -l das * rapid_a_key_tr_dfw.connection.yaml rapid_a_key_tr_ewr.connection.yaml rapid_a_throttling_v2_dfw.connection.yaml rapid_a_throttling_v2_ewr.connection.yaml rapid_cache_hit_dfw.connection.yaml rapid_cache_hit_ewr.connection.yaml

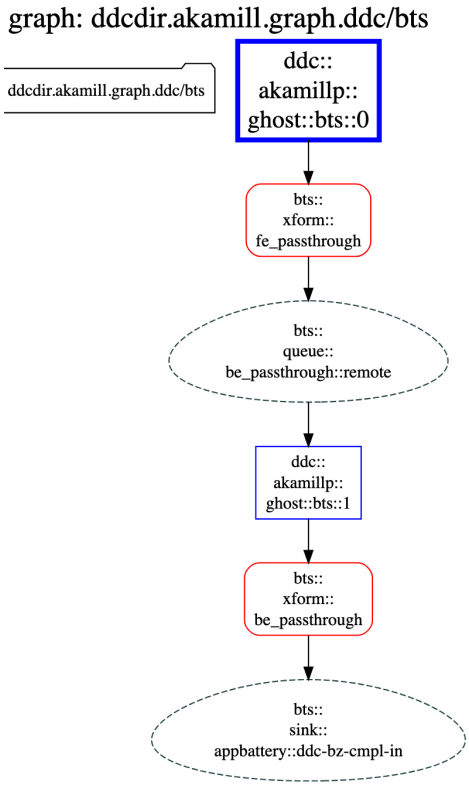
								rapid_cache_miss_dfw.connection.yaml rapid_cache_miss_ewr.connection.yaml rapid_endpoint_q_stats_by_account_v2_fw.connection.yaml rapid_endpoint_q_stats_by_account_v2_wr.connection.yaml rapid_endpoint_q_stats_by_endpoint_v2_ewr.connection.yaml rapid_fwd_turn_fw.connection.yaml rapid_fwd_turn_wr.connection.yaml rapid_jw_invalid_dfw.connection.yaml

								rapid_jv _invalid ewr. connect on.yaml rapid_jv _valid_d w. connect on.yaml rapid_jv _valid_e wr. connect on.yaml rapid_o uth_inva id_dfw. connect on.yaml rapid_o uth_inva id_ewr. connect on.yaml rapid_o uth_vali _dfw. connect on.yaml rapid_o uth_vali _ewr. connect on.yaml rapid_o g_fail_d w. connect on.yaml rapid_o g_fail_e wr. connect on.yaml rapid_o g_fail_v _dfw. connect on.yaml

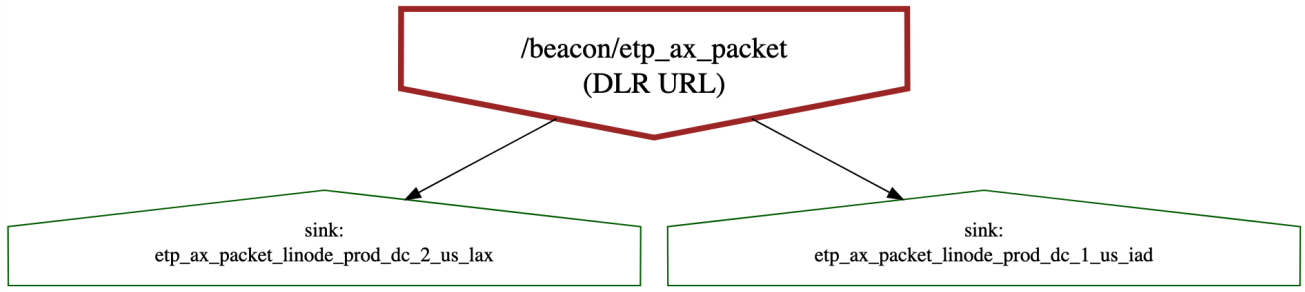
								rapid_o g_fail_v _ewr. connect on.yaml rapid_q ota_der _dfw. connect on.yaml rapid_q ota_sta dfw. connect on.yaml rapid_q ota_sta ewr. connect on.yaml rapid_q ota_sta v2_dfw. connect on.yaml rapid_q ota_sta v2_ewr. connect on.yaml rapid_re q_deny_ dfw. connect on.yaml rapid_re q_deny_ ewr. connect on.yaml rapid_re q_err_d w. connect on.yaml rapid_re q_err_e r. connect on.yaml

								rapid_re q_succ_ dfw. connect on.yaml rapid_re q_succ_ ewr. connect on.yaml rapid_rs _turn_d w. connect on.yaml rapid_rs _turn_e r. connect on.yaml rapid_tf _rsp_df . connect on.yaml rapid_tf _rsp_ev . connect on.yaml waf_exc eptions. connect on.yaml waf_exc eptions_ chicago connect on.yaml

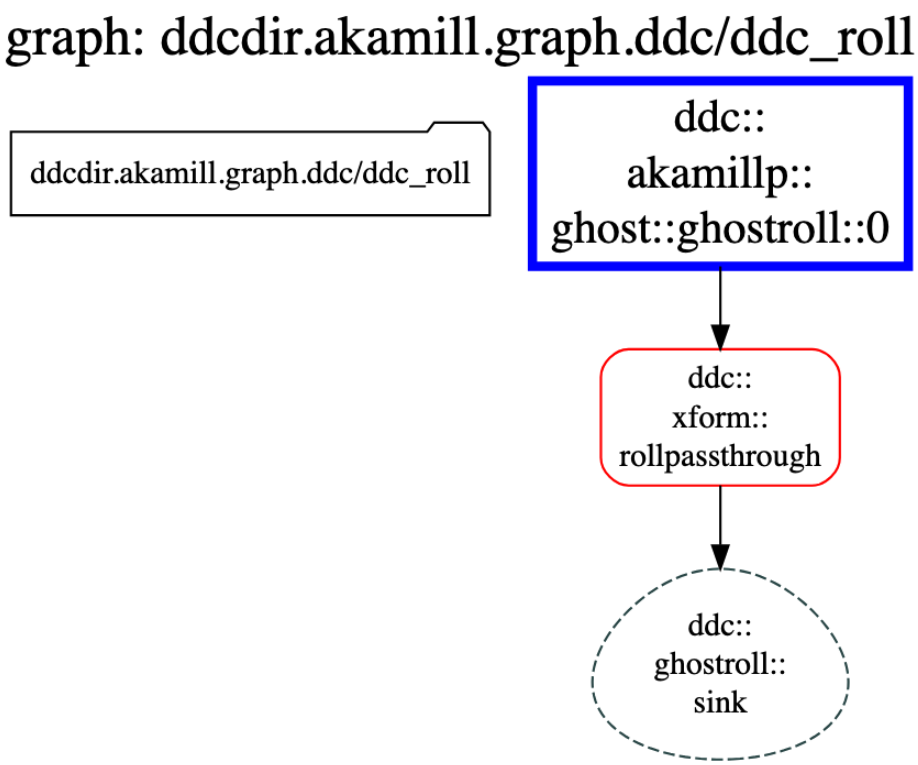
BTS - passthru - flow



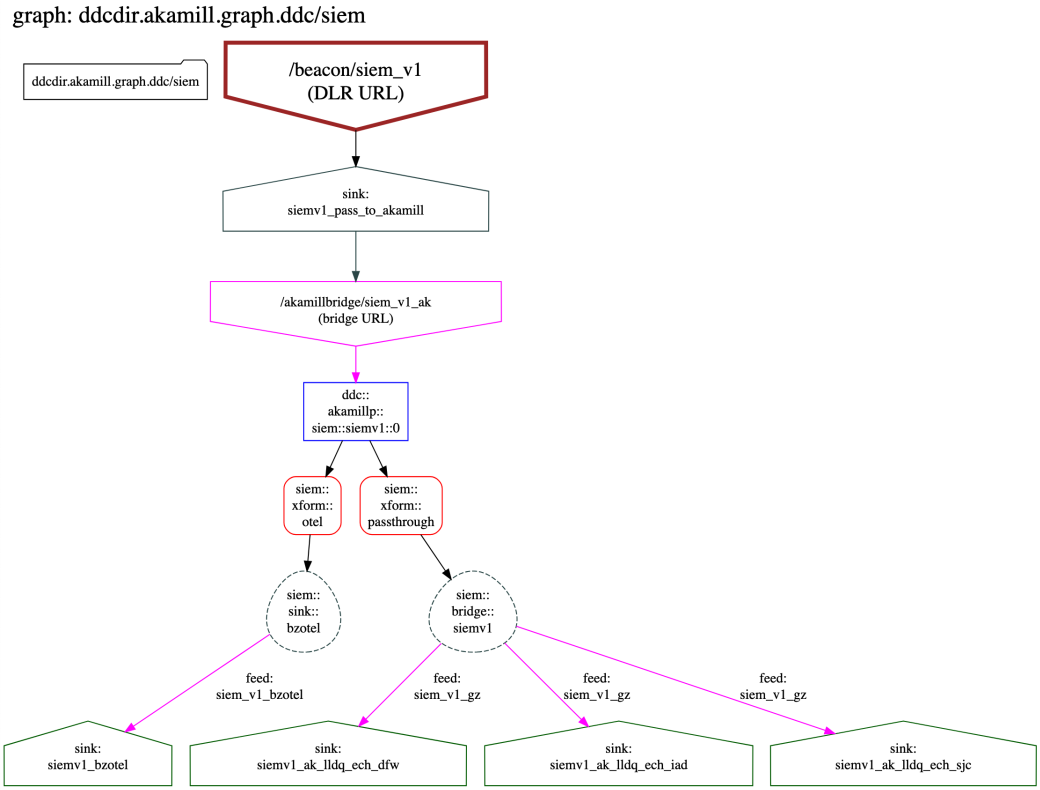
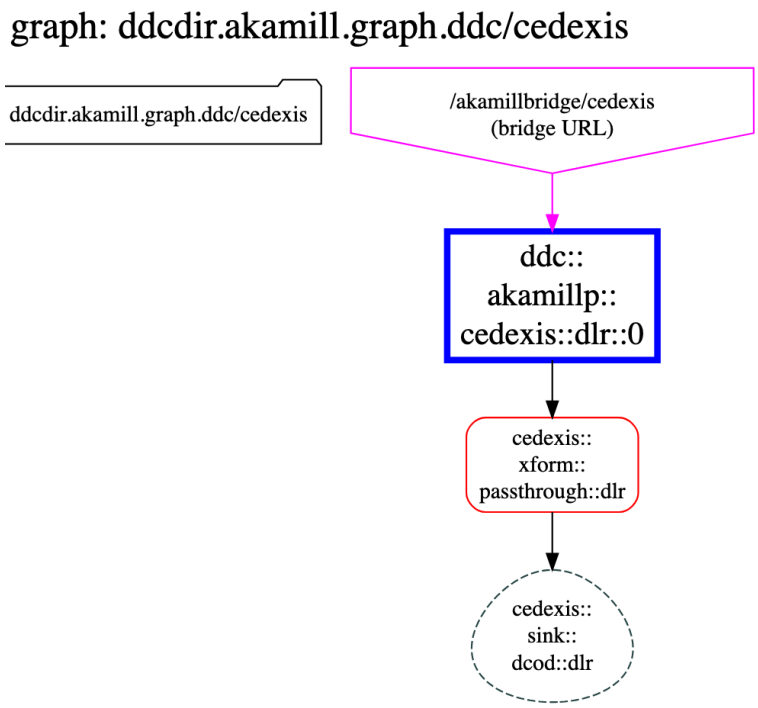
Beacon - ETP AnswerX - flow



Ghost rolls - pass through - flow



Beacon - SIEM - flow



akamillbridge - cedexis DLR passthrough - flow